

A compact PCB based solution for accurately sensing and monitoring air quality in real time.

Team Name AirSense

Program Name & Class : T.Y. Int B.tech. ECE

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Introduction

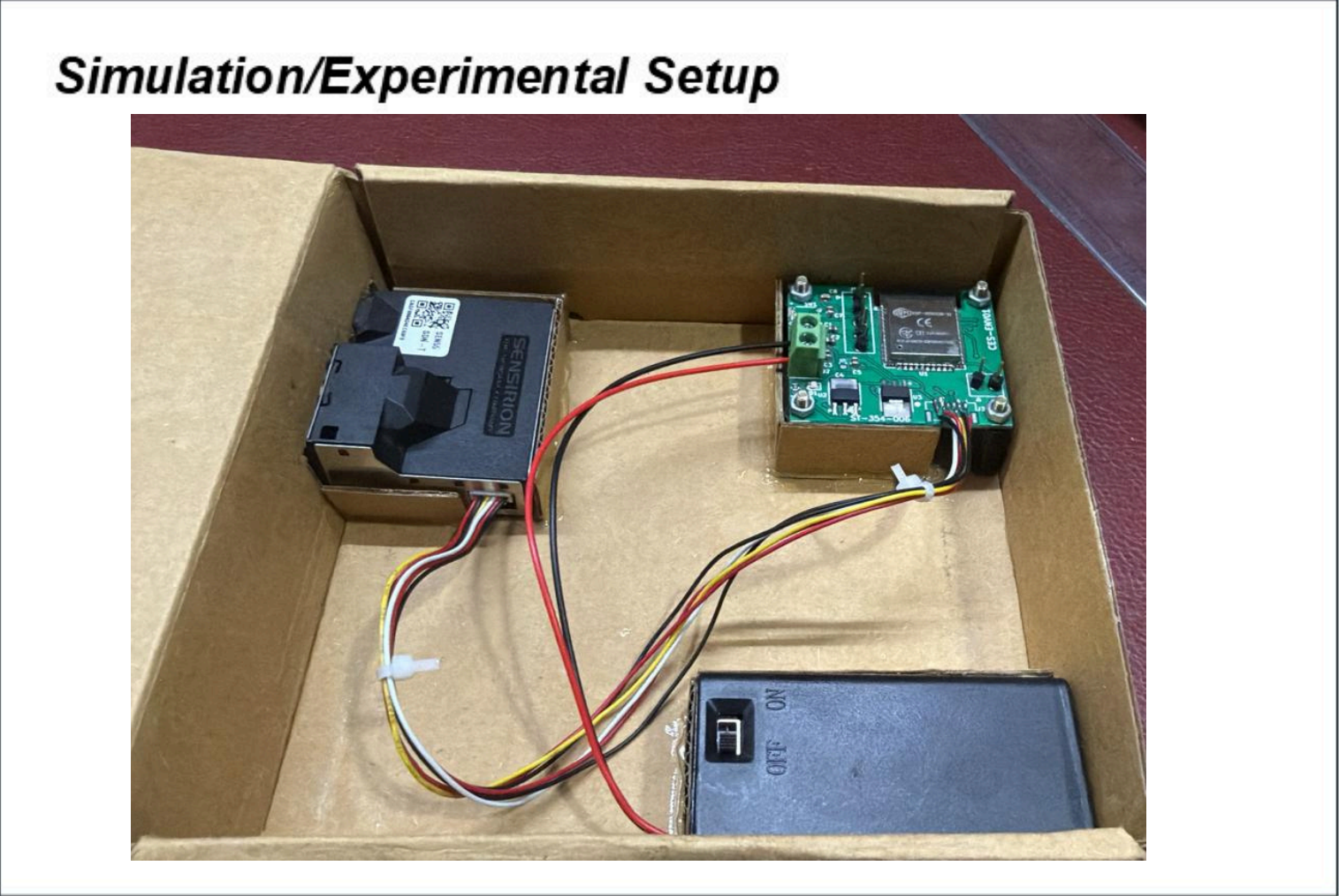
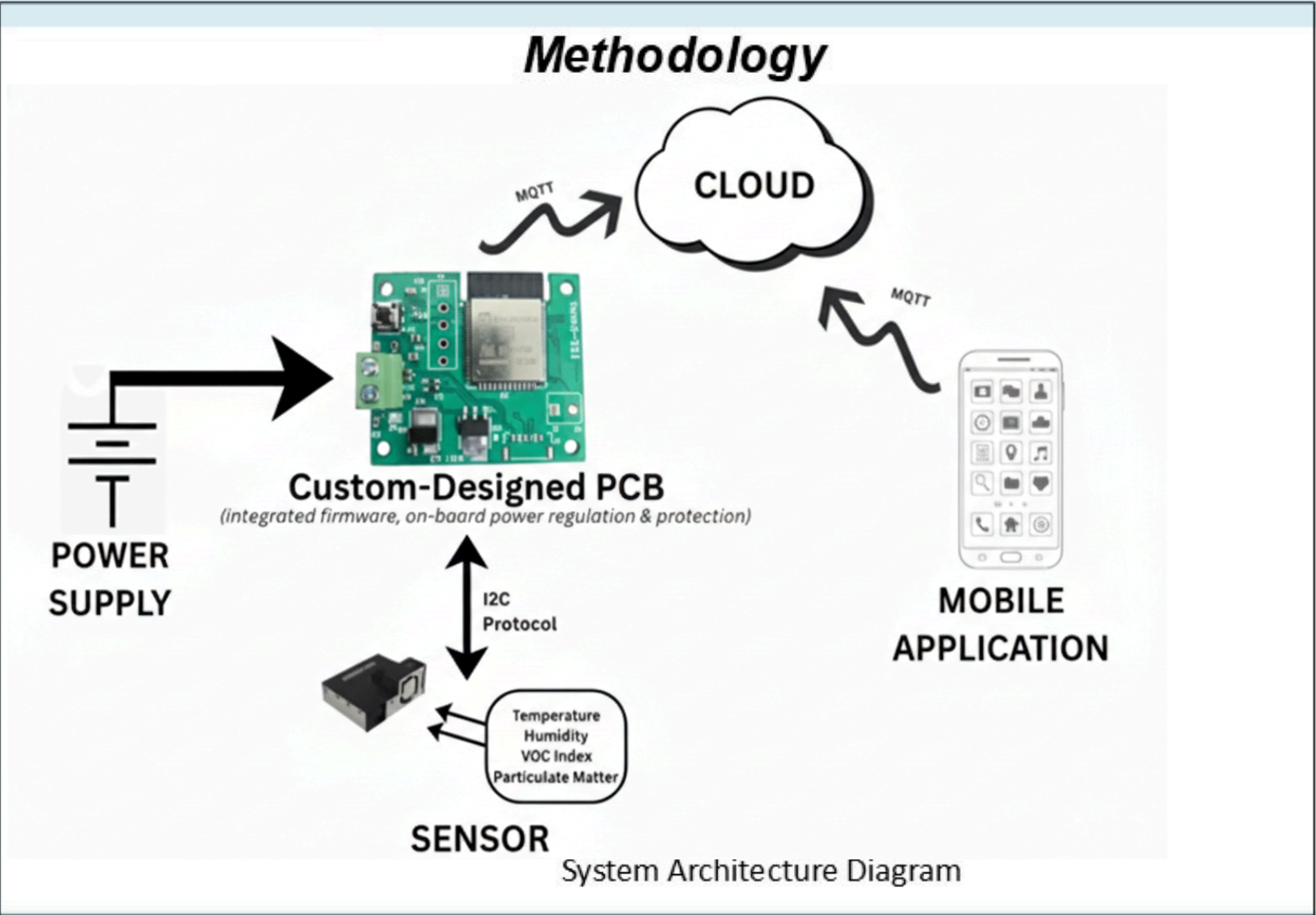
Air pollution has emerged as a significant environmental and public health challenge due to rapid urbanization, industrialization, and increasing vehicular emissions. Long-term exposure to polluted air contributes to respiratory diseases, cardiovascular disorders, and environmental instability. While large-scale smart city monitoring systems exist, they often rely on complex infrastructure and high deployment costs. Therefore, there is a need for a compact, low-cost, and easily deployable air quality monitoring solution suitable for localized indoor and near-outdoor environments. This project presents a PCB-based IoT-enabled air quality monitoring system that integrates sensing, processing, and wireless communication into a single compact device.

Aim

To design and implement a compact PCB-based IoT air quality monitoring system capable of real-time sensing, AQI computation, and wireless data transmission.

Objectives

- To measure particulate matter (PM1.0, PM2.5, PM4.0, PM10)
- To measure temperature and relative humidity
- To measure VOC and NOx indices
- To compute AQI using linear interpolation formula
- To transmit data using MQTT protocol
- To develop a real-time mobile application interface
- To design and manufacture a custom PCB for reliable integration



Results and Discussion(with chart / result table/images)

Env.	AQI	PM1.0 (µg/m³)	PM2.5 (µg/m³)	PM10 (µg/m³)	VOC Index	NOx Index
Ambient Air	145	62	92	145	190	35
Enclosed Box	125	48	78	120	150	28
Box+Agarbatti	190	95	135	210	260	48
Box+5Agarbatti	310	180	280	420	420	75
CrowdedPlace	220	110	165	250	300	60
No crowd	155	70	100	160	200	38

DETAILED METRICS

PM1.0

43.5 µg/m³

PM2.5

47.6 µg/m³

PM4.0

49.2 µg/m³

PM10

50.0 µg/m³

VOC

98 Index

NOX

1 Index

HUMIDITY

42 %

TEMP

25.0 °C

127

AQI

Unhealthy for Sensitive Groups

Good Hazardous

Live • Just now

Summary

A compact IoT-based air quality monitoring system was successfully designed and implemented using SEN55 and ESP32. The system integrates sensing, processing, and wireless communication on a custom PCB and provides real-time environmental data through a mobile application. The platform is suitable for indoor and localized monitoring environments such as homes, offices, and educational institutions. With proper power regulation and calibration, the system provides reliable and scalable air quality assessment.

Important References:

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- SEN55 Datasheet, Sensirion AG, 2023
- ESP32 Technical Reference Manual, Espressif Systems, 2022